AR-11 Mechanical Ventilation, Adult

MEDICAL-DIRECTOR APPROVED APP CONTENT | SOURCE-PDF REPLACEMENT | Reconciled October 17, 2026

Scope: Adult advanced-airway patient requiring mechanical ventilation or an interfacility/home-ventilator patient. Maintain safe established settings when they are known and functioning; use this pathway for a new controlled setup or clinically necessary adjustment.

New controlled setup

1. Confirm	Waveform EtCO2, tube depth, bilateral exam, securement, SpO2, hemodynamics, and analgesia/sedation. Use AR-08 after every ETT or BIAD.
2. Choose mode	Use volume assist-control for a new controlled setup. Do not overwrite stable home or transfer settings without a clinical reason.
3. Set tidal volume	Use 6-8 mL/kg Ideal Body Weight (IBW), based on height and sex - never actual scale weight. Start at 6 mL/kg for ARDS, restrictive physiology, or elevated pressures.
4. Start oxygenation	PEEP 5 cm H2O. FiO2 100% only until a reliable oxygen-saturation measurement is available; then use the lowest effective FiO2 and small, reassessed PEEP changes.
5. Reassess	Synchrony, chest excursion, waveform EtCO2 trend, SpO2, exhalation, hemodynamics, PIP, and plateau pressure when a valid measurement is possible.

Rate and oxygenation branch

Patient / goal	Approved action
Usual adult physiology	Begin 10-12/min. Titrate only after reassessing patient response, waveform EtCO2 trend, and blood-gas information when available.
COPD / asthma	Begin 8-10/min with high inspiratory flow and I:E about 1:4-1:5. Confirm expiratory flow returns to baseline; avoid air trapping and auto-PEEP.
Oxygen target	SpO2 92-98% unless a condition-specific target applies. After ROSC, target SpO2 90-98%; EtCO2 is not a direct substitute for PaCO2.

Alarm or deterioration: If the cause is not immediately clear, disconnect the ventilator, manually ventilate with a BVM, stabilize the patient, and evaluate DOPES/DOTT before restarting or changing settings.

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PRESSURE, SPECIAL-POPULATION, AND TROUBLESHOOTING SAFETY

Pressure and waveform interpretation

Finding	Meaning and response
Plateau pressure	Target below 30 cm H2O. Measure only with an inspiratory hold in a passive patient; an active patient does not provide a reliable plateau measurement.
High PIP + high plateau	Consider reduced compliance: lung disease/ARDS, pulmonary edema, atelectasis, pneumothorax, chest-wall restriction, or increased intra-abdominal pressure. Reduce tidal volume in 1 mL/kg IBW steps to no lower than 4 mL/kg IBW while correcting the cause and obtaining additional direction as needed.
High PIP + normal plateau	Consider airway/circuit resistance: kink, secretions, bronchospasm, biting, narrowed airway, water/filter issue, or auto-PEEP.
Auto-PEEP / stacked breaths	Allow complete exhalation. If unstable or alarm cause is uncertain, disconnect briefly and BVM slowly while reassessing rate, flow, obstruction, and chest excursion.

Special safety branches

- Post-ROSC: use AC-10. Target SpO2 90-98% and, when blood-gas information is available, PaCO2 35-45 mm Hg unless directed otherwise.

- Suspected brain injury: avoid routine hyperventilation. Target normal ventilation unless signs of impending herniation or Medical Control direction require a different plan.

- Severe metabolic acidosis: do not abruptly remove compensatory minute ventilation. Reassess frequently and seek Medical Control or receiving-facility direction before a major reduction in ventilation.

- Use AR-08 for analgesia, sedation, continuous tube confirmation, and post-airway reassessment. Use AR-09 for home-ventilator failure, equipment troubleshooting, and transport preparation.

Do not chase a single number: EtCO2 trends can guide reassessment, but EtCO2 may not equal PaCO2 in poor perfusion or after ROSC. Do not force a numeric EtCO2 target when it worsens air trapping in obstructive disease.

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IDEAL BODY WEIGHT QUICK REFERENCE AND DOCUMENTATION

Ideal Body Weight (IBW) by height and sex

IBW is calculated from height and sex, not actual scale weight. Formula: female = 45.5 + 2.3 x inches over 60; male = 50 + 2.3 x inches over 60. Use the 6 mL/kg end of the range for ARDS, restrictive physiology, or elevated pressures.

Height	Female IBW / 6-8 mL/kg	Male IBW / 6-8 mL/kg
5'0"	45.5 kg / 273-364 mL	50.0 kg / 300-400 mL
5'3"	52.4 kg / 314-419 mL	56.9 kg / 341-455 mL
5'6"	59.3 kg / 356-474 mL	63.8 kg / 383-510 mL
5'9"	66.2 kg / 397-530 mL	70.7 kg / 424-566 mL
6'0"	73.1 kg / 439-585 mL	77.6 kg / 466-621 mL
6'3"	80.0 kg / 480-640 mL	84.5 kg / 507-676 mL
6'6"	86.9 kg / 521-695 mL	91.4 kg / 548-731 mL

DOPES / DOTT emergency check

Tool	Use when ventilation worsens or an alarm persists
DOPES	Displacement; Obstruction; Pneumothorax/pulmonary problem; Equipment failure; Stacked breaths/auto-PEEP.
DOTT	Disconnect or decompress when indicated; Oxygen/BVM; Tube check; Then adjust settings only after the cause is addressed.

Documentation and release control

- Document height, sex used for IBW, initial and changed settings, oxygen/EtCO2 trends, PIP/plateau values when measured, waveform findings, analgesia/sedation, alarms, troubleshooting, patient response, and destination notification.

- This direct source PDF replaces the legacy imported sheet and matches approved Claiborne EMS app content reconciled October 17, 2026.

- Follow agency release governance, training requirements, current medication/device availability, and condition-specific or Medical Control direction when they differ from this pathway.